USART Driver Analysis

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Introduction

USART provides asynchronous serial communication between STM32 and external devices. The custom USART driver allows configuration and communication without using vendor-provided HAL libraries. USART provides asynchronous serial communication between STM32 and external devices. The custom USART driver allows configuration and communication without using vendor-provided HAL libraries.

Driver Architecture

Applications communicate through USART APIs. The driver configures peripheral registers and manages data transmission and reception through TX and RX lines. Applications communicate through USART APIs. The driver configures peripheral registers and manages data transmission and reception through TX and RX lines.

Data Structures

usart_config_t stores baud rate, parity, stop bits, word length, and mode settings. usart_handle_t combines the peripheral base address with configuration data. usart_config_t stores baud rate, parity, stop bits, word length, and mode settings. usart_handle_t combines the peripheral base address with configuration data.

Component	Purpose
Configuration Structure	Stores settings
Handle Structure	Stores peripheral and config
APIs	Interface to application

Register Analysis

SR stores status flags. DR is used for transmitting and receiving data. BRR configures baud rate. CR1, CR2, and CR3 control operating parameters. SR stores status flags. DR is used for transmitting and receiving data. BRR configures baud rate. CR1, CR2, and CR3 control operating parameters.

API Analysis

usart_init() configures the peripheral. usart_send_data() transmits bytes. usart_receive_data() receives data. usart_clock_control() enables peripheral clocks. usart_init() configures the peripheral. usart_send_data() transmits bytes. usart_receive_data() receives data. usart_clock_control() enables peripheral clocks.

Application Example

A UART echo application was implemented. Characters received from a terminal are transmitted back, validating both RX and TX functionality. A UART echo application was implemented. Characters received from a terminal are transmitted back, validating both RX and TX functionality.

Advantages and Conclusion

The driver simplifies serial communication, encourages modular design, and demonstrates understanding of STM32 communication peripherals. The driver simplifies serial communication, encourages modular design, and demonstrates understanding of STM32 communication peripherals.